

Automated Water Purity Monitoring System

Data Acquisition & Signal Processing Project

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Project Overview

This project focuses on the development of an automated water purity monitoring system, designed for real-time analysis of water turbidity.

- **Goal:** To build a low-cost, automated system capable of continuously monitoring water turbidity in real time.
- **Functionality:** The system is engineered to track water clarity trends, facilitating the early detection of potential contamination events.
- **Practical Applications:** Key applications include enhanced monitoring for maintaining optimal aquarium health and broader general water quality surveillance.

Data Acquisition

Our system employs a robust data acquisition pipeline to ensure accurate and reliable water turbidity measurements. The process involves several key stages:

01

Sensor Voltage Readings

The **Raspberry Pi Pico** is responsible for precisely reading the analog voltage output from the turbidity sensor, forming the raw input for our system.

02

Turbidity Value Calibration

Subsequently, a pre-determined **calibration equation** is applied to convert the raw sensor voltage into accurate and meaningful real turbidity values.

03

Timestamping

Each acquired reading is meticulously **timestamped** using an integrated Real-Time Clock (RTC) module, ensuring the chronological integrity of the data.

04

Data Logging

For continuous monitoring, data is logged every **20 minutes to EEPROM**, providing a persistent record of water purity over time.

05

Continuous Collection

This systematic approach facilitated the continuous collection of **100 data points**, allowing for comprehensive trend analysis and anomaly detection.

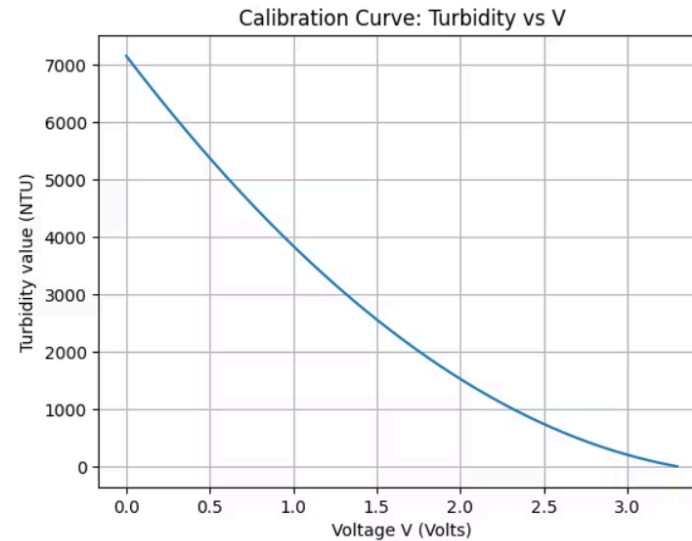
Sensor Calibration

- Calibrated the turbidity sensor using a professional turbidity meter as reference
- Measured voltage output at known turbidity values
- Derived a quadratic calibration equation:

$$T = 496.8 \cdot V^2 - 3804.8 \cdot V + 7154.5$$

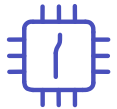
Where T = Turbidity (NTU) and V = Sensor voltage output

The calibration curve illustrates the non-linear relationship between the sensor's voltage output and corresponding turbidity values, providing a graphical representation of the derived quadratic equation.



Hardware Components

Our automated water purity monitoring system relies on a carefully selected set of hardware components to ensure accurate data acquisition and processing:



Raspberry Pi Pico

This serves as the central **microcontroller**, responsible for processing sensor data, managing the system's logic, and communicating with other modules.



Turbidity Sensor with module

The primary sensor for **real-time measurement of water turbidity**, providing crucial data on water clarity and potential contamination.



RTC Module (real-time clock for timestamps)

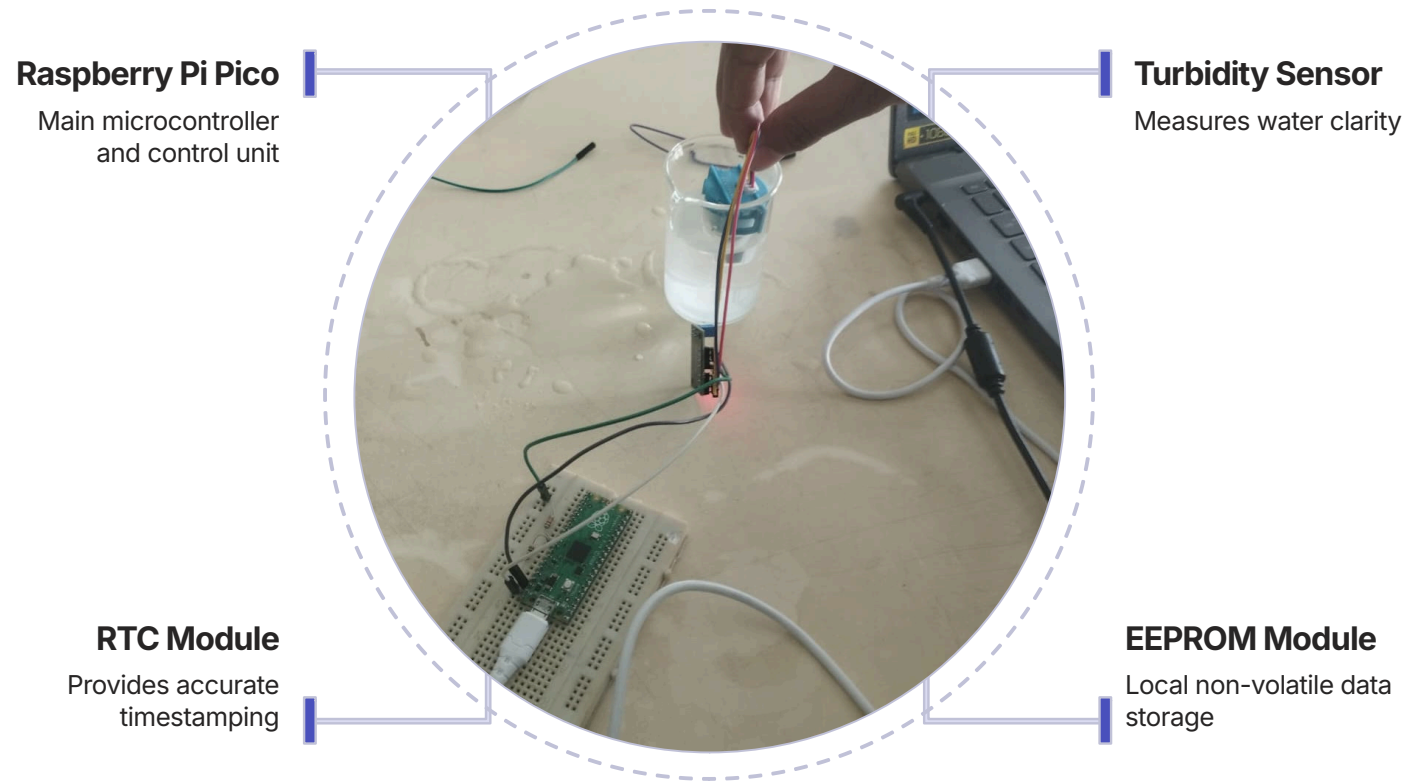
Ensures accurate **timestamping of all collected data**, which is vital for trend analysis and identifying event timelines.



EEPROM module (24c08) for non-volatile data storage

Provides **non-volatile data storage** for configuration settings and logged data, ensuring persistence even during power loss.

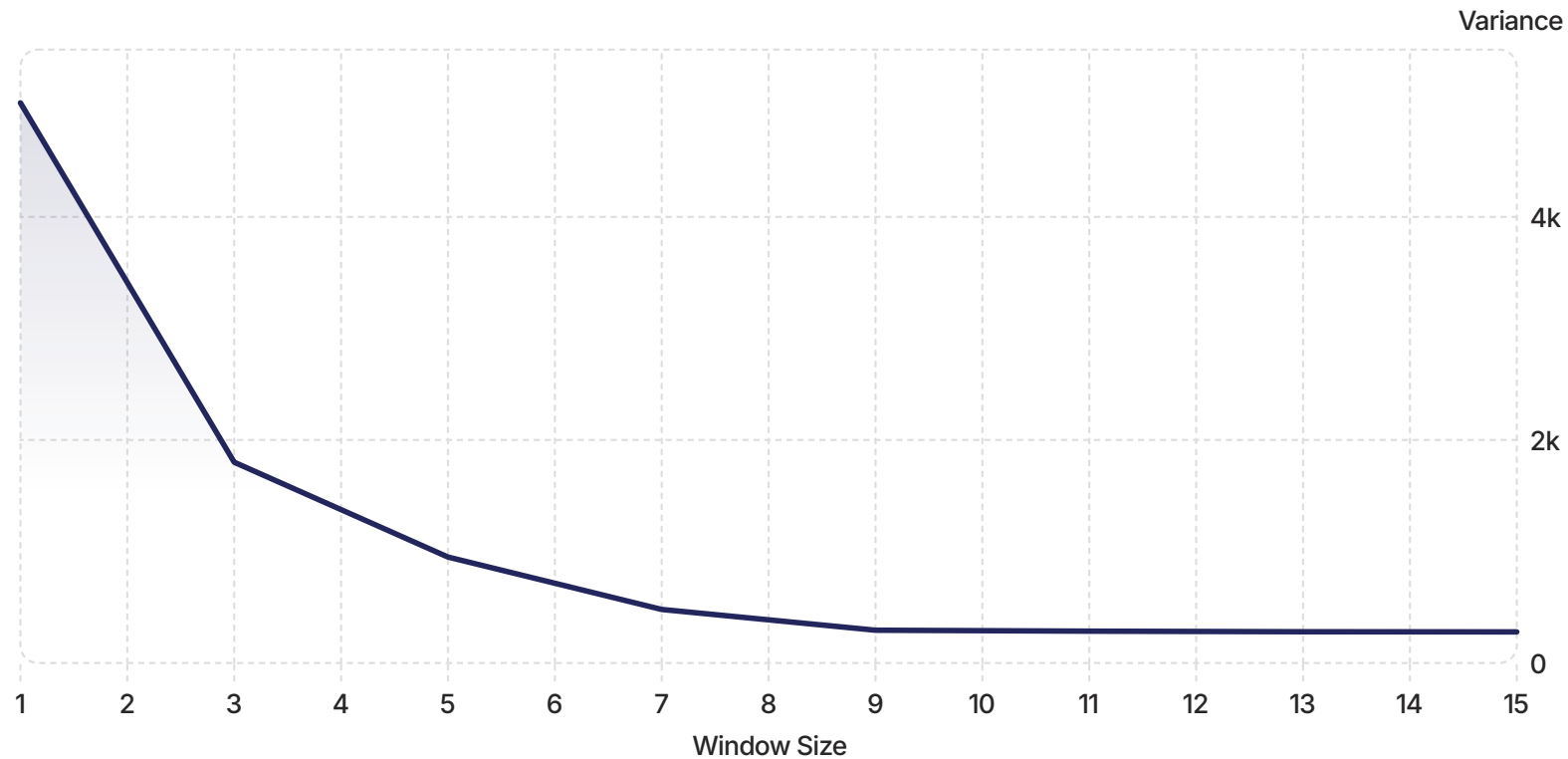
System Architecture



This diagram illustrates the interconnectedness of our system's core hardware. The Raspberry Pi Pico acts as the brain, collecting data from the turbidity sensor, recording timestamps via the RTC module, and storing critical information on the EEPROM module.

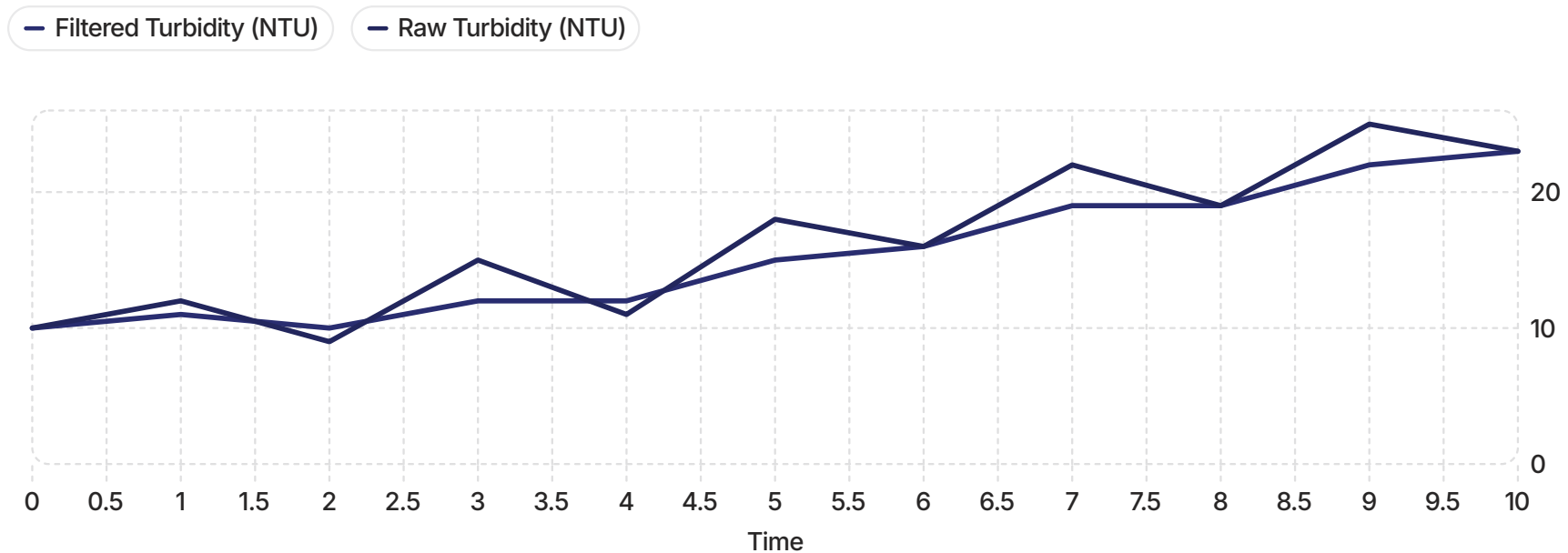
Optimal Filter Selection: Median Filter

After sensitivity analysis, Median Filter with window size 9 was selected as optimal. This selection was made after observing that while variance generally decreases with increasing window size, the improvement becomes negligible beyond a window size of 9, indicating diminishing returns in noise reduction without further compromising signal detail.



The variance vs. window size graph clearly illustrates the diminishing returns in variance reduction after a window size of 9. Beyond this point, increasing the window size offers minimal further benefit in smoothing the data, while potentially introducing undesirable latency or distortion to rapid changes.

The final filtered signal is smooth while preserving the real upward trend in turbidity. This balance is crucial for accurate early detection of contamination while minimizing false alarms from transient noise.



This chart demonstrates the effectiveness of the Median Filter ($w=9$) in producing a smooth output signal. Despite the fluctuations in the raw turbidity data, the filtered signal maintains a clear upward trend, accurately reflecting changes in water purity without being overly affected by transient noise spikes. This ensures reliable and actionable insights for monitoring water quality.

Signal Filtering & Noise Reduction

To ensure accurate and stable turbidity measurements, several signal filtering techniques were evaluated to clean the raw sensor data. The comparison focused on how effectively each method could smooth out noise while preserving essential signal characteristics.

Moving Average (w=5)

This filter calculates the average of a specific number of preceding data points (window size $w=5$), effectively smoothing out short-term fluctuations and reducing random noise.

Exponential Moving Average (EMA, $\alpha=0.25$)

The EMA assigns exponentially decreasing weights to older data points, making it more responsive to recent changes while still providing a smoothed signal. A smoothing factor ($\alpha=0.25$) was used.

Median Filter (w=5)

This non-linear filter replaces each data point with the median value of its neighbors within a window ($w=5$). It is particularly effective at removing salt-and-pepper noise and preserving edge sharpness compared to linear filters.

Hampel Filter (w=5, 3σ)

The Hampel filter is designed to detect and replace outliers in a signal. It identifies points that deviate significantly (e.g., more than 3 standard deviations, 3σ) from the median within a sliding window ($w=5$), replacing them with the median value.

Turbidity Measurements Over Time

This chart displays the raw turbidity measurements collected over a three-hour period, showing data rising from approximately 75 NTU at 19:43 to around 330 NTU by 22:43. The visible spikes and noise in the data indicate real-time fluctuations and potential disturbances within the water.



Conclusion & Future Work

Key Findings

System successfully captures real-time turbidity trends for 100 data points

Median filter ($w=9$) provides the best balance of noise reduction and signal preservation

Demonstrates practical application for water quality monitoring

Future Improvements

Add pH and TDS sensors for comprehensive water quality analysis

Implement threshold alerts for contamination detection

Move to wireless data transmission (Wi-Fi/Bluetooth) for remote monitoring

Expand data logging capacity and analysis capabilities